SafeCharge-RL

Reinforcement learning for EV charging when drivers' departure times are wrong or misreported

Quick guide: the whole project on a few pages, in plain words.

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One sentence. We train an RL policy to share a charging station's limited power among EVs on real data, and test what happens when drivers' reported departure times are wrong or faked.

The problem

- Many cars, one transformer: total power must stay under a **grid limit**.
- Prices change during the day; every driver has a deadline; high-power charging wears batteries.
- The scheduler only sees the departure time the driver **reported**. Real reports are often wrong (about 3 h off on average), and drivers could report early to be served first.

Why RL

- The future (arrivals, prices) is unknown, so we decide step by step.
- Simple rules look at one thing; RL can balance cost, urgency and wear.
- RL has no built-in safety guarantee, so we add a hard safety layer.

What exists and our gap

- Papers cover constrained RL for one EV, stations with random early departures, and rule-based fixes (Ferragut). Products use fixed limits or MPC.
- **Gap:** nobody (to our search) studied a *learned* policy that can be *gamed by misreporting*, on *real* sessions with a *shared hard grid limit*.

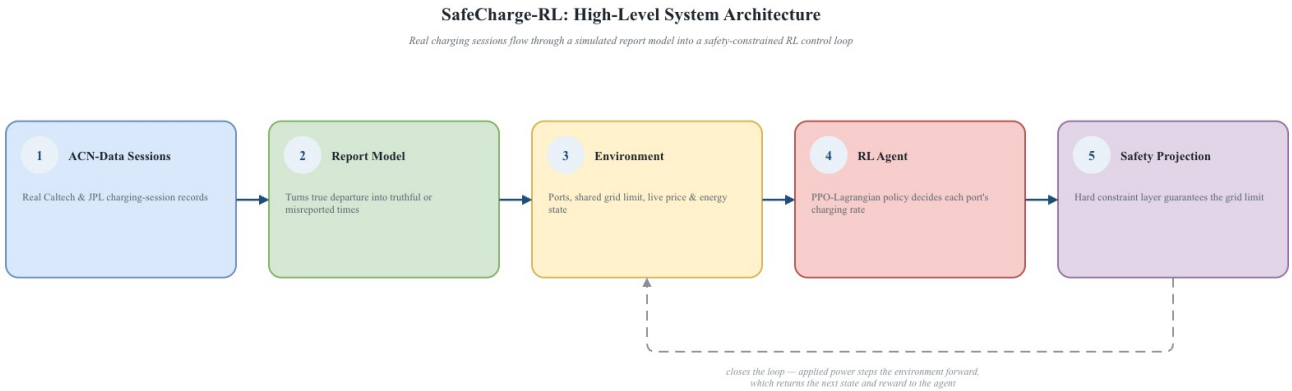
Data

- Real ACN-Data: Caltech 31,424 and JPL 33,638 sessions. One day = 96 steps of 15 min. Port max 6.656 kW. Time-of-use prices (peak $0.267/kWh, 12-18h).
- Train Jan-Aug 2019, validate Sep-Oct 2019, test Nov 2019-Feb 2020 (85 days). JPL and post-COVID used as transfer tests.
- Grid cap = 20% of the typical peak (34.6 kW), so the station is overloaded part of the time.

The RL formulation

Agent / episode	Station controller; one episode = one weekday (96 steps)
State	Per port: occupied, energy needed, time to REPORTED departure, laxity; plus price, time of day, last total power. True departure hidden.
Action	Per port, a fraction (0 to 1) of its max useful power
Reward	Minus electricity cost of the step
Costs (budgets)	Unmet energy at departure (2% budget); battery-wear proxy (share of ports above 80% power)
Safety layer	Subtract the same amount from every request until total power fits the grid limit (hard guarantee)
Algorithm	PPO-Lagrangian + Deep Sets network (works for any number of cars)

Reports tested: truthful (M0), noisy (M2), late (M3), strategic early reports (M4), plus real recorded reports (M1r, extra).



Flow: real sessions, report model, environment, RL agent, safety projection, back to the environment.

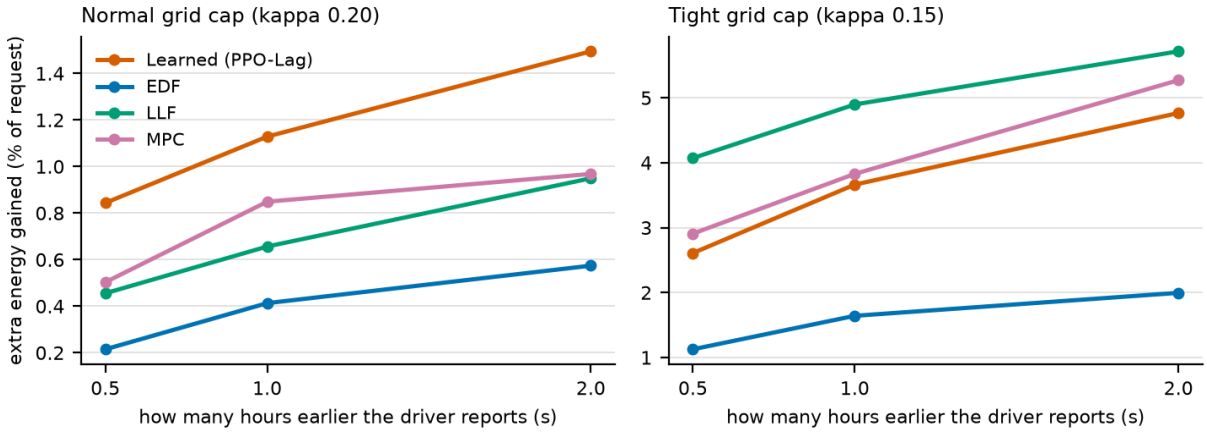
Compared against

- FCFS, EDF (earliest reported departure), LLF (least spare time), MPC (re-plan each step), and the **offline optimum** (best possible schedule knowing true departures).

Results so far (85 test days, 5 seeds, no grid violations anywhere)

Method	Cost $/kWh	Gap to optimum	Unmet energy	Missed-deadline vehicles	High-power ratio
Offline optimum	0.1564	-	0.43%	0.6%	0.79
MPC	0.1649	+5.5%	0.67%	2.5%	0.77
FCFS	0.1708	+10.2%	4.79%	9.1%	0.78
Learned (PPO-Lag)	0.1728	+11.5%	1.32%	7.4%	0.17
EDF	0.1732	+11.7%	1.25%	1.2%	0.77
LLF	0.1738	+12.1%	0.43%	1.7%	0.77

- **Truthful reports:** the learned policy ties EDF; it does not beat the rules. It charges gently (wear proxy about 40% lower).
- **Wrong reports:** learned policy about 2.6x worse, rules about 1.2-1.6x, MPC about 30x.
- **Cheating works:** reporting earlier gets the learned policy's driver extra energy in 93-97% of cases; rules only when crowded. Gain grows with how early and with a tighter cap.
- **Fix:** serving a car only until its declared departure removes the gain for every method, but hurts honest drivers whose reports are early by mistake.
- **Transfer:** fine after COVID at Caltech; fails at JPL (28.6% unmet).



Novel parts

- Reported departure as something a driver may fake, tested on a learned policy with real data.
- Paired-replay measure of misreporting gain.

- Hard projection + budgets for a shared station.
- Comparison with hindsight optimum; test of the Ferragut fix on RL.
- Extra: real driver reports are common and 2-3 h wrong (the '0.2%' in the submitted text is wrong).

Status

Done	Data, simulator, safety layer, baselines, offline optimum, MPC, PPO-Lagrangian (5 seeds), experiments E1, E2, E3, E6, first part of E4, 108 tests passing, code on GitHub
In progress	E4: training with random reports; 6 of 9 runs done, 3 paused (about 1-2 h to finish)
Left	E5 ablation, unprojected RL baseline, SAC-Lagrangian (optional), final report and slides

Honest points: learned policy needs a warm start copied from LLF (not in design); several constants were our choices (all logged in docs/PROJECT_LOG.md); misreporting is simulated.

Map to mid-review slides

1 Title	cover
2 Problem	'The problem', 'Why RL'
3 Existing work	'What exists and our gap'
4 MDP + algorithm + data	'Data', 'The RL formulation'
5 System diagram	the figure on page 2
6 Novel parts	'Novel parts'
7 Demo + next plan	'Results', 'Status'